Exercises Week 7 (**with solutions**)

Advanced Machine Learning (02460)
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Exercise 7.1 (Metric of randomly projected manifold) Consider the manifold spanned by the mapping $f : \mathbb{R}^d \to \mathbb{R}^{2D}$ defined as

$$h(\mathbf{x}) = \begin{pmatrix} f(\mathbf{x}) \\ \sigma(\mathbf{x}). \end{pmatrix} \tag{1}$$

1. *Derive the pullback metric of this manifold.*

Solution 7.1 The Jacobian of h is

$$\mathbf{J}_h = \begin{pmatrix} \mathbf{J}_f \\ \mathbf{J}_\sigma \end{pmatrix}. \tag{2}$$

The metric is then

$$\mathbf{G}_h = \mathbf{J}_h^\intercal \mathbf{J} = \mathbf{J}_f^\intercal \mathbf{J}_f + \mathbf{J}_\sigma^\intercal \mathbf{J}_\sigma. \tag{3}$$

Exercise 7.2 (Choice of parametrization) Consider two normal distributions

$$\mathcal{N}_1 = \mathcal{N}(\mu_1, \sigma_1^2) \qquad \text{and} \qquad \mathcal{N}_2 = \mathcal{N}(\mu_2, \sigma_2^2), \tag{4}$$

where

$$\mu_1 = 0, \quad \mu_2 = 1, \quad \sigma_1 = 1, \quad \sigma_2 = 2. \tag{5}$$

1. *Compute the Euclidean distance between these distributions in the following parametrizations*

$$\theta = (\mu, \sigma) \tag{6}$$
$$\theta = (\mu, \sigma^2) \tag{7}$$
$$\theta = (\mu, \sigma^{-2}) \tag{8}$$
$$\theta = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2} \right). \tag{9}$$

2. *Does one parametrization seem more natural (or better) than another (this question is open-ended)?*

3. *Find parametrization of normal distributions such that the associated pullback metric corresponds to that of the randomly projected manifold (Eq. 13.8 in the LMLG book).*

Solution 7.2 For the first question, we evaluate the Euclidean distances directly

$$\left\| \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\| = \sqrt{1+1} = \sqrt{2} \tag{10}$$

$$\left\| \begin{pmatrix} 0 \\ 1^2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2^2 \end{pmatrix} \right\| = \sqrt{1+3^2} = \sqrt{10} \tag{11}$$

$$\left\| \begin{pmatrix} 0 \\ 1^{-2} \end{pmatrix} - \begin{pmatrix} 1 \\ 2^{-2} \end{pmatrix} \right\| = \sqrt{1+(3/4)^2} = \frac{\sqrt{25}}{\sqrt{16}} = \frac{5}{4} \tag{12}$$

$$\left\| \begin{pmatrix} 0 \\ -1/2 \end{pmatrix} - \begin{pmatrix} 1/4 \\ -\frac{1}{8} \end{pmatrix} \right\| = \sqrt{\left(\frac{2}{8}\right)^2 + \left(\frac{3}{8}\right)^2} = \sqrt{\frac{4}{64} + \frac{9}{64}} = \sqrt{\frac{13}{64}}. \tag{13}$$

Clearly choice of parametrization matters.

The second question is intended to be open-ended, so we provide no solution here.

For the third question, we note that the parametrization $\theta = (\mu, \sigma)$ gives a pull-back metric corresponding Eq. 12.8.

Exercise 7.3 (Bias-free ReLU networks) Consider a neural network consisting of layers

$$f_l(\mathbf{x}) = \mathrm{ReLU}(\theta_l \, \mathbf{x}), \tag{14}$$

such that the joint network is

$$f(\mathbf{x}) = (f_L \circ \ldots f_1)(\mathbf{x}) = f_L(\ldots f_2(f_1(\mathbf{x}))). \tag{15}$$

1. Show that two parameter vectors θ and θ' produce the same function f if

$$\sum_{l=1}^{L} \log \theta_l = \sum_{l=1}^{L} \log \theta'_l. \tag{16}$$

Solution 7.3 First, note that our function can be written as

$$f(\mathbf{x}) = \left(\prod_{l=1}^{L} \theta_l \right) \mathrm{ReLU}(\mathbf{x}). \tag{17}$$

We then see that θ and θ' produce identical functions if

$$\prod_{l=1}^{L} \theta_l = \prod_{l=1}^{L} \theta'_l. \tag{18}$$

Taking the logarithm on both sides yields the desired result.

Exercise 7.4 (ReLU geodesics) Consider the neural network

$$f(\mathbf{x}) = \theta_2 \, \mathrm{ReLU}(\theta_1 \, \mathbf{x}) \tag{19}$$

and observational data

$$\begin{pmatrix} \mathbf{x}_1 \\ \mathbf{y}_1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} \mathbf{x}_2 \\ \mathbf{y}_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{20}$$

This data can perfectly fit with the parameters $\theta_1 = \theta_2 = 1$.

1. Show that $\theta_1 = 7$ and $\theta_2 = 1/7$ provides an equally good fit to the data.
2. Derive the finite-sample estimate of the metric $\mathbf{G}_\theta = \mathbb{E}[\mathbf{J}_\theta(\mathbf{x})^\intercal \mathbf{J}_\theta(\mathbf{x})]$ given by

$$\bar{\mathbf{G}}_\theta = \frac{1}{2}\Big(\mathbf{J}_\theta(\mathbf{x}_1)^\intercal \mathbf{J}_\theta(\mathbf{x}_1) + \mathbf{J}_\theta(\mathbf{x}_2)^\intercal \mathbf{J}_\theta(\mathbf{x}_2) \Big). \tag{21}$$

3. Write a computer program to compute the geodesic between parameters

$$\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 21 \\ 3/7 \end{pmatrix}. \tag{22}$$

Solution 7.4 The answer to the first question follows directly from Eq. 14.4.

To answer the second question, we first note that the Jacobian is given by

$$\mathbf{J}_\theta(\mathbf{x}) = \begin{pmatrix} \theta_2 \\ \theta_1 \end{pmatrix}^\intercal \mathrm{ReLU}(\mathbf{x}). \tag{23}$$

The finite sample metric then becomes

$$\begin{aligned} \bar{\mathbf{G}}_\theta = \frac{1}{2}\Bigg(& \begin{pmatrix} \theta_2 \\ \theta_1 \end{pmatrix} \begin{pmatrix} \theta_2 \\ \theta_1 \end{pmatrix}^\intercal \mathrm{ReLU}(\mathbf{x}_1) \\ & + \begin{pmatrix} \theta_2 \\ \theta_1 \end{pmatrix} \begin{pmatrix} \theta_2 \\ \theta_1 \end{pmatrix}^\intercal \mathrm{ReLU}(\mathbf{x}_2) \Bigg), \end{aligned} \tag{24}$$

which, for the given data, reduces to

$$\bar{\mathbf{G}}_\theta = \frac{1}{2} \begin{pmatrix} \theta_2 \\ \theta_1 \end{pmatrix} \begin{pmatrix} \theta_2 \\ \theta_1 \end{pmatrix}^\intercal. \tag{25}$$

Exercise 7.5 (Hyperbolic tangent networks) Consider the neural network

$$f(\mathbf{x}) = \theta_2 \tanh(\theta_1 \mathbf{x}) \tag{26}$$

and observational data

$$\begin{pmatrix} \mathbf{x}_1 \\ \mathbf{y}_1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} \mathbf{x}_2 \\ \mathbf{y}_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{27}$$

1. Derive the finite-sample estimate of the metric $\mathbf{G}_\theta = \mathbb{E}[\mathbf{J}_\theta(\mathbf{x})^\intercal \mathbf{J}_\theta(\mathbf{x})]$ given by

$$\bar{\mathbf{G}}_\theta = \frac{1}{2}\Big(\mathbf{J}_\theta(\mathbf{x}_1)^\intercal \mathbf{J}_\theta(\mathbf{x}_1) + \mathbf{J}_\theta(\mathbf{x}_2)^\intercal \mathbf{J}_\theta(\mathbf{x}_2)\Big). \tag{28}$$

   It may be helpful to recall that the derivative of $\tanh(\mathbf{x})$ is $1 - \tanh^2(\mathbf{x})$.

2. Write a computer program to compute the geodesic between parameters

$$\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 21 \\ 3/7 \end{pmatrix}. \tag{29}$$

Solution 7.5 To answer the first question, we note that the Jacobian is given by

$$\mathbf{J}_\theta(\mathbf{x}) = \begin{pmatrix} \theta_2 \mathbf{x} \left(1 - \tanh^2(\theta_1 \mathbf{x}\right) \\ \tanh(\theta_1 \mathbf{x}) \end{pmatrix}^\intercal \tag{30}$$

Evaluated at the given data, this Jacobian is (recall $\tanh(-\mathbf{x}) = -\tanh(\mathbf{x})$)

$$\mathbf{J}_\theta(-1) = \begin{pmatrix} -\theta_2 \left(1 - \tanh^2(-\theta_1 \right) \\ \tanh(-\theta_1) \end{pmatrix}^\intercal \tag{31}$$

$$= -\begin{pmatrix} \theta_2 \left(1 - \tanh^2(\theta_1 \right) \\ \tanh(\theta_1) \end{pmatrix}^\intercal \tag{32}$$

$$\mathbf{J}_\theta(1) = \begin{pmatrix} \theta_2 \left(1 - \tanh^2(\theta_1 \right) \\ \tanh(\theta_1) \end{pmatrix}^\intercal \tag{33}$$

such that $\mathbf{J}_\theta(-1) = -\mathbf{J}_\theta(1)$.

The finite sample metric then becomes

$$\begin{aligned} \bar{\mathbf{G}}_\theta &= \mathbf{J}_\theta(1)^\intercal \mathbf{J}_\theta(1) \\ &= \begin{pmatrix} \theta_2^2 \left(1 - \tanh^2(\theta_1 \right)^2 & \tanh(\theta_1)\, \theta_2 \left(1 - \tanh^2(\theta_1 \right) \\ \tanh(\theta_1)\, \theta_2 \left(1 - \tanh^2(\theta_1 \right) & \tanh^2(\theta_1) \end{pmatrix}. \end{aligned} \tag{34}$$